MOHAMED SATHAK ENGINEERING COLLEGE:: KILAKARAI-623806

(An Autonomous Institution)

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DEPARTMENT: ECE Semester/Year:III/VI

SUB CODE / NAME: ET3491/EMBEDDED SYSTEMS &

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PART-A (10 x 2 = 20 Marks)

1. **Accumulator Function in 8051**:
   Stores intermediate results during arithmetic and logic operations.
2. **Addressing Modes in 8051**:
   Immediate, Register, Direct, Indirect.
3. **Stages in Embedded System Design**:
   Requirement analysis → Specification → Architecture design → Component selection → Coding → Testing.
4. **Supervisor vs User Mode (ARM)**:
   Supervisor mode has full access to system resources, while user mode has restricted access.
5. **Preemptive Real-time OS**:
   An OS where high-priority tasks preempt low-priority ones to meet deadlines.
6. **Multitasking with Example**:
   Running multiple tasks (e.g., motor control + data logging) simultaneously on a microcontroller.
7. **IoT Protocols for Industrial Communication**:
   MQTT, CoAP, Modbus, OPC UA.
8. **Key Role of YANG in IoT Management**:
   YANG models configuration and state data, enabling structured data management.
9. **Hardware Components in Raspberry Pi Board**:
   SoC, RAM, GPIO, HDMI, USB, SD Card slot, WiFi/Bluetooth module.
10. **Role of Python in IoT on Raspberry Pi**:
    Easy-to-use scripting for automation, data handling, and sensor interfacing.

PART-B (5 x 13 = 65 Marks)

11. a) 8051 Microcontroller Architecture

- **Introduction** **(2 Marks)**
- **Block Diagram Explanation** **(7 Marks)**
- **Components** **(4 Marks)**
  - ALU, Accumulator, B Register
  - Program Counter & Stack Pointer
  - RAM & ROM
  - Ports (P0 to P3)
  - Timers/Counters
  - Serial Communication Block
  - Interrupt Control

11. b) Timers and Interrupts in 8051 with Example Program

Introduction **(2** Marks**)**

Timers and interrupts are essential features in 8051 for precise timing and event-driven execution.

Types of Timers **(4** Marks**)**

- Timer 0 and Timer 1 (16-bit)
- Modes: Mode 0 (13-bit), Mode 1 (16-bit), Mode 2 (8-bit auto-reload)

Interrupts in 8051 **(4** Marks**)**

- External Interrupts: INT0 and INT1
- Timer Interrupts: TF0 and TF1
- Serial Interrupt

Application Example **(3** Marks**)**

LED blinking every 1 ms using timer interrupt.

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12. a) Embedded System Design with Model Train Case Study (6*2+1=13 Marks)

- **Requirement Analysis**
- **Specification Development**
- **Architecture Selection**
- **Hardware/Software Partitioning**
- **Prototyping & Implementation**
- **Testing and Debugging**
- **Design Constraints & Safety Aspects**

12. b) ARM Instruction Set Preliminaries and Performance Impact

(3*4+1=13 Marks)

Instruction Types

- **Data Processing Instructions** (ADD, SUB, MOV)
- **Load/Store Instructions** (LDR, STR)
- **Branch Instructions** (B, BL)
- **Program Status Register Instructions**

Thumb vs ARM Instructions

- **ARM**: 32-bit, powerful, more registers.
- **Thumb**: 16-bit, smaller size, suitable for memory-limited systems.

Impact on Performance

- **Thumb mode** saves memory but has limited instruction set.
- **Conditional Execution** reduces branching.
- **Pipeline Execution** improves instruction throughput.

Performance Justification

Choose ARM for speed, Thumb for space-constrained applications.

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13. a) Real-time System Structure **(6*2+1=13 Marks)**

- **Definition of Real-time System**
- **Task Assignment Mechanism**
- **Task Scheduling Techniques**
- **Multitasking with Priorities**
- **Use of Preemptive/Non-preemptive Schedulers**
- **Diagram Representation**
- **Advantages and Use Cases**

13. b) Priority-Based Scheduling and Interprocess Communication in RTOS

Priority-Based Scheduling **(4 Marks)**

- **Fixed Priority Scheduling (FPS)**
- **Rate Monotonic Scheduling (RMS)**
- **Earliest Deadline First (EDF)**

Preemptive Mechanism **(4 Marks)**

- High-priority task preempts lower ones.
- Guarantees deterministic behavior.

Interprocess Communication (IPC) **(4 Marks)**

- **Shared Memory**
- **Message Queues**
- **Semaphores and Mutexes**

Importance **(1 Mark)**

- Ensures reliability, avoids deadlock/starvation.
- Maintains real-time constraints.

Example Use Case

Sensor data acquisition system where critical sensor tasks are assigned higher priority.

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14. a) IoT Reference Architecture **(3*4+1=13** Marks**)**

- **Definition and Purpose**
- **Layered Architecture:**
  - Perception Layer
  - Network Layer
  - Middleware Layer
  - Application Layer
  - Business Layer
- **Scalability and Interoperability**
- **Use in Various IoT Applications**

14. b) NETCONF and YANG in IoT System Management (3*4+1=13 **Marks**)

NETCONF Overview

- Network Configuration Protocol (RFC 6241)
- Used to install, manipulate, and delete configuration of network devices.

YANG Overview

- Data modeling language used by NETCONF
- Defines structure of configuration and state data

Benefits in IoT

- Scalable management of IoT devices
- Enables automation and consistency
- Secure and structured data handling

Use in Large-Scale Deployment

- Telecom, Industrial IoT use NETCONF/YANG for managing thousands of devices efficiently.

15. a) Sensor Interface with Raspberry Pi using Python (6*2+1=13 Marks)

- **Overview**
- **Sensor Selection (e.g., DHT11, LM35)**
- **Connection and Pin Configuration**
- **Python Code Sample**
- **Data Acquisition and Processing**
- **Display and Logging**
- **Real-world Use Case**

15. b) IoT Implementation in Home Automation vs Smart Cities

Home Automation (4 **Marks**)

- **Scope**: Localized to a single house
- **Technologies**: Wi-Fi, Zigbee, Bluetooth
- **Applications**: Smart lighting, thermostats, appliances
- **Challenges**: User-friendliness, privacy

Smart Cities **(4 Marks)**

- **Scope**: City-wide systems
- **Technologies**: LoRaWAN, 5G, NB-IoT
- **Applications**: Smart traffic, waste management, surveillance
- **Challenges**: Scalability, interoperability, data analytics

Comparison Summary **(5 Marks)**

Feature	Home Automation	Smart Cities
Network	Local Wi-Fi/Zigbee	Wide-area LPWAN/5G
Complexity	Low	High
Devices	10s	1000s to Millions
Data Handling	Basic	Big Data / AI Needed

PART-C (1 x 15 = 15 Marks)

16. a) Temperature Monitoring System using 8051 (5*3=15 Marks)

- **Required Components**
  - 8051 Microcontroller
  - LM35 Temperature Sensor
  - Serial Communication (UART)

- **Algorithm Steps**
- **Assembly/C Embedded Code Sample**
- **Serial Data Transmission**
- **System Design Justification**
    - Cost-effectiveness
    - Simplicity
    - Real-time Performance

16. b) Embedded Program in ARM Assembly for I/O and Exception Handling

(5*3=15 Marks)

Objective

Create a program that performs basic input/output and handles exceptions using ARM assembly.

Program Structure

- **I/O Operations**: Reading a switch and toggling an LED
- **Exception Handling**: SWI (Software Interrupt) or IRQ

Sample Code Snippet

```
assembly
CopyEdit
MOV R0, #0x40021000   ; GPIO port base address
LDR R1, [R0, #0x10]    ; Read input pin
CMP R1, #0x01
BEQ toggle_LED

toggle_LED:
   LDR R2, [R0, #0x14]
   EOR R2, R2, #0x01
   STR R2, [R0, #0x14]

SWI #0x0            ; Software interrupt (simulate exception)
```

Exception Handler

Defined in the vector table – performs logging or resets system.

Performance Evaluation

- **Low latency I/O**
- **Efficient use of registers**
- **Minimal overhead**
- Chosen instructions minimize memory access and enhance speed.